Exercises: Tree-based Classification

Exercise 1 (Feature selection): Consider the following training examples for a binary classification problem, where we want to predict the value of Class attribute (C0 or C1?)

Table 4.1. Data set for Exercise 2.

Customer ID	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

(a) Consider Gender, CarType, and ShirtSize, which attribute is the best for predicting Class(C0/C1)? Justify your answer in terms of Information-Gain.

Answer: Class distribution: 10 C0, 10 C1 -> Entropy(S) = 1.000. Information Gain Gender: 0.029, CarType: 0.620, ShirtSize: 0.012. CarType is the best attribute: it has by far the largest Information Gain, because every “Sports” customer is C0 and almost every “Luxury” customer is C1, so CarType alone nearly separates the two classes, while Gender and ShirtSize each split the data close to 50/50 in every branch and barely reduce uncertainty.

(b) Explain why Customer ID should not be used as the attribute test condition even though it has the lowest Entropy.

Answer: Customer ID is unique to every record, so splitting on it creates 20 branches with exactly one customer each — every branch is “pure” (entropy = 0), which makes its

Information Gain look maximal. But this does not generalize: a brand-new customer has an ID never seen during training, so the tree would have no branch for them. High-cardinality identifier attributes can look best purely because they memorize the training set, not because they capture a real, repeatable pattern, so they should be excluded from candidate split attributes.

Exercise 2 (Building decision tree): Given the following training data: Buy Computer data

ID	age	income	student	credit_rating	buy
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31...40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31...40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31...40	medium	no	excellent	yes
13	31...40	high	yes	fair	yes
14	>40	medium	no	excellent	no

a) Step by step, build a decision tree using Information-Gain as splitting criteria

Answer: Base entropy: 14 records, 9 “yes” / 5 “no” -> Entropy(S) = 0.940. Information Gain at the root age: 0.246 (highest), income: 0.029, student: 0.151, credit_rating: 0.048, so age is the root split, with branches age ≤ 30 (2 yes, 3 no), age 31–40 (4 yes, 0 no, already pure), age > 40 (3 yes, 2 no). Within age ≤ 30, student perfectly separates the remaining records (student = no -> all “no”; student = yes -> all “yes”), so student is the next split there. Within age > 40, credit_rating perfectly separates the remaining records (fair -> all “yes”; excellent -> all “no”), so credit_rating is the next split there. Resulting tree:

- age ≤ 30 -> student?
  - student = no -> buy = NO
  - student = yes -> buy = YES
- age 31...40 -> buy = YES (leaf, already pure)
- age > 40 -> credit_rating?
  - credit_rating = fair -> buy = YES
  - credit_rating = excellent -> buy = NO

b) Use the constructed tree to predict the class of the following new example: age<=30, income=medium, student=yes, credit-rating=fair.

Answer: age ≤ 30 -> student = yes -> buy = YES. (income and credit_rating are not needed on this path, since age ≤ 30 is already fully resolved by student alone.) Predicted class: buy = YES.